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Ice maker with stand pipe drain

Abstract

In an ice maker with a sump below an ice formation device and a pump for recirculating water from the sump to the ice formation device, water can drain through a stand pipe drain. A drain fitting is disposed in relation to a stand pipe such that the drain fitting's upper end is disposed above the top of the stand pipe, its lower end defines a water inlet adjacent the bottom of the sump, and its perimeter wall extends 360° about the stand pipe to define a drain passage extending from the water inlet to the top of the stand pipe. The upper end of the drain fitting defines a siphon release opening. The stand pipe drain can also include a valve for adjusting the drain between an overflow drain configuration and a siphon drain configuration.

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Background/Summary

FIELD

(1) This disclosure generally pertains to an ice maker with a stand pipe drain.

BACKGROUND

(2) Certain dedicated ice maker appliances employ a water inlet that selectively imparts supply water into a sump so that water can be circulated from the sump to an ice formation device. It is known to provide a drain in the sump to prevent accidental overflow.

SUMMARY

(3) In one aspect, an ice maker comprises an ice formation device. A sump is below the ice formation device. The sump has a bottom wall. A pump recirculates water from the sump to the ice

formation device so that the water forms as ice on the ice formation device. A stand pipe extends upward from the bottom wall of the sump and has an open upper end portion. The stand pipe is configured so that water can flow into the stand pipe through the open upper end portion to drain from the sump. A drain fitting has a lower end portion, an upper end portion opposite the lower end portion, and a perimeter wall extending from the lower end portion to the upper end portion. The drain fitting is disposed in relation to the stand pipe such that the upper end portion of the drain fitting is disposed above the open upper end portion of the stand pipe, the lower end portion of the drain fitting defines a water inlet adjacent the bottom wall of the sump, and the perimeter wall extends 360° about the stand pipe such that the stand pipe and the drain fitting define a drain passage extending from the water inlet to the open upper end portion of the stand pipe. The upper end portion of the drain fitting defines a siphon release opening configured to provide fluid communication between the drain passage and an area outside the drain fitting.

(4) In another aspect, a method of using an ice maker comprises using a stand pipe in a sump of the ice maker as an overflow drain. The ice maker is adjusted to use the stand pipe as siphon drain.

(5) In another aspect, an ice maker comprises an ice formation device. A sump is below the ice formation device. A stand pipe drain is in the sump. The stand pipe drain is selectively adjustable between an overflow drain configuration in which the stand pipe drain is configured to drain water from the sump to an overflow water level and a siphon drain configuration in which the stand pipe drain is configured to drain water from the sump to a water level less than the overflow water level.

(6) Other aspects will be in part apparent and in part pointed out hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of an ice maker in the scope of this disclosure;
- (2) FIG. 2 is a schematic block diagram of a control system of the ice maker;
- (3) FIG. 3 is an enlarged schematic illustration of part of the ice maker, showing a stand pipe drain thereof in an overflow drain configuration;
- (4) FIG. 4 is an enlarged schematic illustration similar to FIG. 3 but showing the stand pipe drain in a siphon drain configuration;
- (5) FIG. 5 is a perspective of a subassembly of the ice maker including a sump, a water pump, and a drain fitting of the stand pipe drain;
- (6) FIG. 6 is a top plan view of the subassembly of FIG. 5;
- (7) FIG. 7 is an exploded perspective of the sump and the drain fitting;
- (8) FIG. 8 is a cross section taken in the plane of line 8-8 of FIG. 6;
- (9) FIG. 9 is a cross section taken in the plane of line 9-9 of FIG. 6;
- (10) FIG. 10 is an enlarged view of a portion of FIG. 9;
- (11) FIG. 11 is a perspective of the drain fitting;
- (12) FIG. 12 is an elevation of the drain fitting;
- (13) FIG. 13 is a cross-section taken in the plane of line 13-13 of FIG. 12;
- (14) FIG. 14 is a cross section taken in the plane of line 14-14 of FIG. 12;
- (15) FIG. 15 is an enlarged fragmentary cross section of a portion of the subassembly of FIG. 5 at the same plane of the drain fitting as in FIG. 14; and
- (16) FIG. 16 is an enlarged schematic illustration similar to FIG. 3 but showing a prior art stand pipe drain.

(17) Corresponding parts are given corresponding reference characters throughout the drawings.

DETAILED DESCRIPTION

(18) Referring to FIG. 1, an exemplary embodiment of an ice maker in the scope of this disclosure is shown schematically at reference number 103. In one aspect, this disclosure pertains to a drain

system for the ice maker **103**. As will be explained in further detail below, the drain system of the illustrated ice maker can be configured to draw out impurities from the water so that relatively pure water is used to make ice. The illustrated drain system can also be selectively adjustable to operate as either an overflow drain or a siphon drain.

(19) An ice maker in the scope of this disclosure may broadly comprise an ice formation device in which water can form into ice, a water system for directing water onto the ice formation device, and a refrigeration system configured to cool the ice formation device to a temperature at which at least some of the liquid water present in the ice formation device will freeze into ice. In the illustrated embodiment, the ice maker **103** is a vertical spray ice maker of the type which has a horizontally oriented freeze plate **110** that constitutes the ice formation device. The horizontal freeze plate **110** defines a plurality of ice molds **111** that open downward to receive water **S** sprayed upward from below. Those skilled in the art will recognize that this type of ice maker is used to make very hard, clear ice. Other types of ice makers such as batch ice makers with vertically oriented freeze plates are also contemplated to be in the scope of this disclosure. Batch ice makers with vertically oriented freeze plates differ from the illustrated ice maker in that the freeze plate extends in a generally vertical plane, with a water distributor above the vertical freeze plate so that water flows down the freeze plate during ice making cycles.

(20) The refrigeration system of the ice maker **103** includes a compressor **112**, a heat rejecting heat exchanger **114**, a refrigerant expansion device **118** for lowering the temperature and pressure of the refrigerant, an evaporator **120** along the top side of the freeze plate **110**, and a hot gas valve **124**. The compressor **112** can be a fixed speed compressor or a variable speed compressor to provide a broader range of control possibilities. The compressor **112** cycles a form of refrigerant through the condenser **114**, expansion device **118**, evaporator **120**, and the hot gas valve **124**, via refrigerant lines.

(21) As shown, the heat rejecting heat exchanger **114** may comprise a condenser for condensing compressed refrigerant vapor discharged from the compressor **112**. In other embodiments, e.g., in refrigeration systems that utilize carbon dioxide refrigerants where the heat of rejection is trans-critical, the heat rejecting heat exchanger is able to reject heat from the refrigerant without condensing the refrigerant. In certain embodiments that utilize a gaseous cooling medium (e.g., air) to provide condenser cooling, a condenser fan **115** may be positioned to blow the gaseous cooling medium across the condenser **114**. The condenser fan **115** can be a fixed speed fan or a variable speed fan to provide a broader range of control possibilities.

(22) Hot gas valve **124** is configured to be selectively opened and closed to control freezing and harvesting of ice with the refrigeration system. During freezing, the hot gas valve **124** is closed to direct warm refrigerant vapor to the condenser **114**. During ice harvest, the hot gas valve **124** is configured to open to direct warm refrigerant from the compressor **112** directly to the evaporator **120** to demold and harvest ice cubes from the freeze plate **110**.

(23) The refrigerant expansion device **118** can be of any suitable type, including a capillary tube, a thermostatic expansion valve, or an electronic expansion valve. In certain embodiments, where the refrigerant expansion device **118** is a thermostatic expansion valve or an electronic expansion valve, the ice maker **103** may also include a temperature sensor (not shown) placed at the outlet of the evaporator **120** to control the refrigerant expansion device **118**. In other embodiments, where the refrigerant expansion device **118** is an electronic expansion valve, the ice maker **103** may also include a pressure sensor (not shown) placed at the outlet of the evaporator **120** to control the refrigerant expansion device **118** as is known in the art.

(24) Referring still to FIG. **1**, a water system of the illustrated ice maker **103** includes a sump **130**, a water pump **132**, a water line **134** (broadly, passaging), and a water level sensor **136** (shown schematically in FIG. **2**). Water level sensor **136** can be any suitable type of water level sensor for signaling when water level in the sump reaches certain control thresholds. Examples of water level sensors commonly used in ice makers of this type include float sensors and pneumatic sensors. The

water pump **132** could be a fixed speed pump or a variable speed pump to provide a broader range of control possibilities.

(25) The water system of the ice maker **103** further includes a water supply line **138** and a water inlet valve **140** for filling the sump **130** with water from a water source (e.g., a municipal water utility). The illustrated water system further includes a stand pipe drain, generally indicated at reference number **141**. The stand pipe drain **141** is connected to a drain line **142** for draining water from the sump **130**. As will be explained in further detail below, the stand pipe drain **141** includes a drain valve **144** for selectively controlling drainage through the stand pipe drain.

(26) The sump **130** is positioned below the freeze plate **110** to catch water falling from the freeze plate. The water line **134** fluidly connects the water pump **132** to a sprayer **146** below the freeze plate **110**. The sprayer **146** is configured to spray liquid water S upward to the freeze plate **110**. A slanted ice chute **147** is located between the sprayer **146** and the downward facing freeze plate **110**. The ice chute **147** comprises a grill or other porous structure that allows spray S and falling liquid water to pass through the chute. But the ice chute **147** is configured to block pieces of frozen ice I from falling through the chute. Instead, harvested ice pieces I land on the chute **147** and slide forward, falling off of the bottom of the chute into the ice bin **104** below.

(27) During an ice batch production cycle, the pump **132** is configured to pump water through the water line **134** and through the sprayer **146**. The liquid S is sprayed upward past the chute **147** into the molds **111** of the freeze plate **110**. Some of the water freezes in the molds **111** and unfrozen liquid water falls off of the freeze plate **110**, past the chute **147** and the sprayer **146**, into the sump **130** where it can be recirculated by the water pump **132**. This water cycle continues to progressively chill the liquid water that is recirculating until a sufficient amount of the water freezes as ice in the molds **111**. At that point the refrigeration system opens the hot gas valve **124** to heat the freeze plate **110**, melting the ice I until it demolds, falls onto the chute **147**, and slides off the chute into the ice bin **104**.

(28) Referring to FIG. 2, the ice maker **103** includes a controller **160**. The controller **160** includes at least one processor **162** for controlling the operation of the ice maker **103**, e.g., for controlling at least one of the refrigeration system and the water system. The processor **162** may include a non-transitory processor-readable medium storing code representing instructions to cause the processor to perform a process. The processor **162** may be, for example, a commercially available microprocessor, an application-specific integrated circuit (ASIC) or a combination of ASICs, which are designed to achieve one or more specific functions, or enable one or more specific devices or applications. In certain embodiments, the controller **160** may be an analog or digital circuit, or a combination of multiple circuits. The controller **160** may also include one or more memory components **164** for storing data in a form retrievable by the controller. The controller **160** can store data in or retrieve data from the one or more memory components **164**.

(29) In various embodiments, the controller **160** may also comprise input/output (I/O) components to communicate with and/or control the various components of ice maker **103**. In certain embodiments, for example, the controller **160** may receive inputs such as, for example, one or more indications, signals, messages, commands, data, and/or any other information, from the water level sensor **136**, a harvest sensor **166** for determining when ice has been harvested, an electrical power source (not shown), an ice level sensor **141** for detecting the level of ice in the bin **104** (FIG. 1), and/or a variety of sensors and/or switches including, but not limited to, pressure transducers, temperature sensors, acoustic sensors, etc. In various embodiments, based on those inputs and predefined control instructions stored in the memory components **164**, the controller **160** controls the ice maker **103** by outputting control signals to controllable output components such as the compressor **112**, the condenser fan **115**, the refrigerant expansion device **118**, the hot gas valve **124**, the water inlet valve **140**, the drain valve **144**, and/or the water pump **132**. Such control signals may include one or more indications, signals, messages, commands, data, and/or any other information to such components.

(30) The illustrated controller **160** is also operatively connected to a user interface device **165** comprising inputs (e.g., buttons, knobs, a capacitive touchscreen, or the like) through which a user can make commands to the controller and indicators (e.g., a display, a light panel, or the like) for providing indications of information related to the ice maker **103**. Accordingly, the user interface device **165** provides an interface for local interaction with the ice maker **103**. Although not shown, it is to be understood that the ice maker **103** can comprise a network interface device (e.g., a wireless transceiver, a wired Ethernet card, etc.) to provide a remote interface through which an operator can interact remotely with the ice maker.

(31) The remainder of this disclosure focuses on exemplary features of the stand pipe drain **141** of ice maker. For purposes of comparison, FIG. **16** schematically illustrates an ice maker **1103** with a prior art stand pipe drain **1141**. The prior art stand pipe drain **1141** comprises a stand pipe **1205** that extends upward from the bottom wall of the sump **1130**. During use, anytime water in the sump **1130** rises above the top of the stand pipe **1205** the water above the stand pipe **1205** flows into the stand pipe drain **1141** through the open top end of the stand pipe **1205**. The inventor recognizes that the water in sump has a temperature gradient along the height of the sump, with colder, less pure water toward the bottom and warmer, cleaner water toward the top. Thus, when water drains through the prior art stand pipe drain **1141**, it tends to be clean, warm water tends that is discharged, causing impurities to settle and become progressively more concentrated in the colder water at the bottom of the sump over time. As a result, when water is drained from the stand pipe drain **1141**, it is the cleaner, warmer water at the top of the sump that flows into the stand pipe **1205**—not the impure water toward the bottom of the sump **1130**. Furthermore, the water pump **1132** draws the cold, relatively impure water from the bottom of the sump so that the ice maker **1103** makes less pure ice.

(32) Referring now to FIGS. **3** and **4**, in contrast to the stand pipe drain **1141** of the prior art, the stand pipe drain **141** of the present disclosure is configured to draw in water from the bottom of the sump **130**. This enables the ice maker **103** of the present disclosure to drain the relatively impure water at the bottom of the sump while leaving the purer warmer water in the sump. Another optional feature of the of the stand pipe drain **141** of the present disclosure is that it is selectively adjustable between (i) an overflow drain configuration (FIG. **3**) for draining water from the sump **130** to an overflow water level OWL and (ii) a siphon drain configuration (FIG. **4**) for draining water from the sump to a water level less than the overflow water level. The stand pipe drain **141** is configured to drain water through an opening **201** in the bottom wall **203** of the sump **130**.

(33) The stand pipe drain **141** comprises a stand pipe **205** that extends upward from the bottom wall of the sump and a drain fitting **210** that fits over the stand pipe. The stand pipe **205** has a vertical center axis VA centered on the center of the drain opening **201**. The stand pipe **205** has a lower end portion and an upper end portion spaced apart along the vertical axis VA. The lower end portion of the stand pipe **205** is sealingly engaged with the bottom wall **203** of the sump **130**. For instance, in one or more embodiments, the stand pipe **205** and the bottom wall **203** of the sump **130** are integrally formed from a single piece of molded material so that there is a seamless connection of the stand pipe to the bottom wall. The stand pipe **205** has a pipe wall that extends 360° circumferentially about the drain opening **201**. The upper end portion of the stand pipe **205** is open and defines the overflow water level OWL. When water in the sump **130** rises above the overflow water level OWL, it will flow into the open upper end portion of the stand pipe **205** and drain through the drain opening **201**. As explained in further detail below, when the stand pipe drain **141** is in the overflow drain configuration (FIG. **3**), water will drain to the overflow water level OWL and then stop draining. In the siphon drain configuration (FIG. **4**), the stand pipe drain **141** forms a bell siphon so that essentially all of the water in the sump **130** can drain through the stand pipe **205**.

(34) The bell siphon draining process will now be briefly described. In the siphon drain configuration of FIG. **4**, as water fills the sump **130**, the top surface of the water is exposed to atmospheric pressure outside and inside the drain fitting **210**. Outside the drain fitting **210**, the

water is exposed to atmosphere through the open top of sump **130**. Inside the drain fitting **210**, the top of the water is exposed to atmosphere through the empty stand pipe **205**. As the water rises above the overflow water level OWL, it forms a seal above the stand pipe **205** inside the drain fitting **210**, closing off the drain fitting's connection to atmospheric pressure. Then, as the water falls through the stand pipe **205**, it creates a vacuum at the top of the drain fitting **210**, drawing water from the sump **130** up the drain fitting and out through the stand pipe. The column of water in the stand pipe **205** and drain passage **142** continues to pull a vacuum at the top of the stand pipe. As long as the stand pipe drain **141** is maintained in the siphon drain configuration, it will continue to siphon water from the sump **130** until the water level falls below the inlet openings **214** of the fitting **210**.

(35) The drain fitting **210** has a lower end portion and an upper end portion spaced apart along the vertical axis VA. The drain fitting **210** has a perimeter wall **212** extending from the lower end portion to the upper end portion. The drain fitting **210** is disposed in relation to the stand pipe **205** such that the upper end portion of the drain fitting is above the open upper end portion of the stand pipe, the lower end portion of the drain fitting defines a water inlet **214** adjacent the bottom wall **203** of the sump **130**, and the perimeter wall **212** extends 360° circumferentially about the stand pipe in relation to the vertical axis VA such that the stand pipe and the drain fitting define a drain passage **216** extending from the water inlet to the open upper end portion of the stand pipe. The upper end portion of the drain fitting **210** defines a siphon release opening **218** configured to provide fluid communication between the drain passage **216** and an area outside the drain fitting.

(36) Referring to FIGS. 5-10, an exemplary embodiment of a subassembly for an ice maker that includes a sump **130**, a stand pipe **205**, and a drain fitting **210** is generally indicated at reference number **300**. In these drawings, the drain valve **144** of the stand pipe drain **141** is omitted to show the features of and relationships between exemplary embodiments of the stand pipe **205**, the drain fitting **210**, and the sump **130** more clearly. As shown, in the illustrated embodiment, the stand pipe **205** is a substantially cylindrical pipe extending along the vertical axis VA from the bottom sump wall **203**. The perimeter wall **212** of the drain fitting **210** is likewise cylindrical but has a larger diameter than the stand pipe **205**. As a result, when the drain fitting **210** is secured to the stand pipe **205**, the inner surface of the fitting perimeter wall **212** is spaced apart radially outward from the outer surface of the stand pipe **205** by one or more gaps, which define the drain passage **216**.

(37) The drain fitting **210** is configured to be pressed downward onto the stand pipe **205** (see FIG. 7). The lower end portion of the drain fitting **210** is open and the upper end portion of the drain fitting defines a lip **220** around the siphon release opening. **218** The stand pipe **205** has a height H1 (FIG. 10) extending from the bottom wall **203** to the open upper end portion along the vertical axis VA, and the lip **220** is spaced apart from the bottom edge of the fitting **210** by a height H2 (FIG. 2) that is greater than the height H1. Hence, when the drain fitting **210** is installed on the stand pipe **205**, the bottom edge of the fitting **210** engages the bottom wall **203** of the sump **130** and there is a heightwise gap **221** between the upper end portion of the stand pipe and the lip **220**. This heightwise gap **221** allows water above the overflow water level OWL to flow into the open upper end portion of the stand pipe **205**. In addition, the heightwise gap **221** provides the necessary space to form a siphon vacuum chamber above the stand pipe **205** when the stand pipe drain **141** is operating in the siphon drain configuration.

(38) Referring to FIGS. 10-15, the drain fitting **210** comprises a plurality of internal rails **222** circumferentially spaced apart about the perimeter wall **212**. The internal rails **222** are configured to secure the drain fitting **210** on the stand pipe **205**. In the illustrated embodiment, each of the internal rails has a T-shaped cross sectional shape (see FIG. 14). The outer section of the T-shaped rails **222** from the base of the 'T', and the inner section of the T-shaped rails **222** form the two outwardly extending arms at the top of the 'T'. The inner arm sections of the T-shaped rails have concavely curved inner surfaces to match the curved outer surface of the stand pipe **205**. When the drain fitting **210** is pressed onto the stand pipe **205**, the inner surfaces of the T-shaped rails **222**

engage the outer surface of the stand pipe to support the drain fitting on the stand pipe. The circumferentially spaced internal rails **222** define circumferential gaps **224** between them. These gaps **224** define vertical portions of the drain passage that extend upward from the inlet **214** to the upper end portion of the stand pipe **205**. Each internal rail **222** extends along the vertical axis VA from a lower end portion at the lower end portion of the drain fitting **210** to an upper end at the upper end portion of the drain fitting (e.g., to an upper end portion that meets the upper lip **220**). In one or more embodiments, the lower end portion of each internal rail **222** includes a skewed end surface **226** angled to extend inward as it extends upward.

(39) The lower end portion of the illustrated fitting **210** includes a plurality of circumferentially spaced apart inlet notches **228**. The internal rails **222** are circumferentially interleaved between the inlet notches **228** such that there is one rail between each adjacent pair of inlet notches. The inlet notches **228** have open bottom ends. Between the notches **228**, the lower end portion of the fitting **210** is configured to contact the bottom wall **203** of the sump **220** such that the inlet notches define the water inlet **214** (see FIG. **10**). This configuration causes the stand pipe drain **141** to draw water into the drain from the bottom of the sump **130** where impurities tend to be concentrated. As a result of this configuration, the drain **141** can be used periodically to flush some of the water with relatively high concentrations of impurities from the sump **130** so that water with less impurities can be added to the sump and the ice produced by the ice maker maintains high clarity.

(40) Referring again to FIGS. **3** and **4**, the drain valve **144** is configured to selectively open and close the siphon release opening **218** at the upper end portion of the drain fitting **210**. The drain valve **144** comprises a valve member **230** and a valve actuator **232** (e.g., an electric solenoid) configured to selectively move the valve member between an opened position (FIG. **3**) and a closed position (FIG. **4**). More particularly, the illustrated valve actuator **232** is configured to raise the valve member **230** to the opened position and lower the valve member to the closed position. The valve member **230** is configured to sealingly engage the upper end portion of the drain fitting **210** in the closed position to close the siphon release opening **218**. Suitably, the valve member **230** makes an airtight seal with the upper end portion of the drain fitting **210** capable of holding a vacuum in the heightwise gap **221** (FIG. **10**) between the upper end portion of the stand pipe **205** and the lip **220** of the drain fitting **210**.

(41) An exemplary method of using the ice maker **103** will now be briefly described. As explained above, the controller **160** is configured to direct the ice maker **103** to conduct ice making cycles in which the water system and refrigeration system work in concert to form ice **I** in the freeze plate **110** and then demold the ice to harvest it in the bin **104**. In each ice making cycle, the controller **160** will open the water inlet valve **140** to fill the sump **130** to a defined starting level, then run the water pump **132** while using the refrigeration system to chill the freeze plate **110**. During this stage, the sprayer **146** is spraying liquid water **S** into the molds **111**. The freeze plate **110** chills the water so that some of the water freezes in the molds **111** and the remaining water falls back into the sump **130** at a reduced temperature. The vertical spraying continues until the controller **160** determines sufficient ice has formed in the molds **111**. Then the controller **160** opens the hot gas valve **124** to heat the freeze plate **110** until the ice demolds and falls down the chute **147** into the ice bin **104**. This cycle repeats for as long as there is demand for ice.

(42) As ice making cycles are conducted, the controller **160** is configured to flush some or all of the water from the sump **130** through the stand pipe drain **141** to remove impurities. During normal use, the stand pipe drain **141** is in the overflow configuration shown of FIG. **3**. Every n cycles (wherein $n \geq 1$), the controller **160** can conduct a partial flush. The controller **160** keeps the stand pipe drain **141** in the overflow drain configuration in FIG. **3**. At the beginning of the ice making cycle, the controller opens the water inlet valve **140** to fill the sump **130** to a water level greater than the overflow water level OWL and keeps the water inlet valve **140** open for a period of time before closing the valve. This causes the stand pipe drain **141** to drain the amount of water added in excess of the overflow water level OWL. More particularly, the stand pipe drain **140** intakes water

from the bottom of the sump **130** through the water inlets **214** and drains the water taken in through the stand pipe **205**. As explained above, impurities tend to be concentrated in the water at the lower portion of the sump **130**, so the partial flush beneficially flushes relatively impure lower sump water while maintaining a large amount of relatively pure chilled upper sump water in the sump for the next ice making cycle.

(43) On some occasions, the controller **160** is configured to conduct a full drain. For example, in one or more embodiments, the controller **160** is configured to conduct a full drain every m cycles, wherein $m > n$. In some embodiments, the controller **160** is configured to conduct a full drain after receiving a signal from the ice level sensor **141** that the ice bin **104** is full of ice. For example, the controller **160** receives a signal that the ice bin **140** is full of ice, directs the refrigeration system and water system to stop making ice, and conducts a full drain operation as explained below.

(44) During each full drain operation (also called a siphon drain cycle), the controller is configured to direct the drain valve **144** to close and seal the siphon release opening **218**. The controller **144** also directs the water inlet valve **140** to fill the sump to a drain starting level DL greater than the overflow water level OWL. When the siphon release opening **218** is closed, as water above the overflow water level OWL begins to drain through the stand pipe **205**, it creates a vacuum in the gap **221** (FIG. **10**) between the upper end portion of the stand pipe and the now-closed upper end portion of the drain fitting **210**. This vacuum will continue to draw water from the sump **130** through the inlets **214** into the stand pipe **205** until the vacuum is broken. The vacuum can be broken at any time by opening the drain valve **144** to reconfigure the drain **141** in the overflow drain configuration. But if the drain valve **144** is not opened, the vacuum in the upper end portion of the drain fitting **210** will draw water out of the sump **130** until the water level falls below the inlet **214**—i.e., until the sump is essentially empty.

(45) Accordingly, it can be seen that the inventor has provided an ice maker **103** with a stand pipe drain **141** that can be used selectively as an overflow drain (by keeping the siphon release opening **218** open) and a siphon drain (by closing the siphon release opening **218**). The inventor believes that this multi-purpose stand pipe drain can provide enhanced drainage capabilities without substantially increasing the manufacturing complexity or cost over conventional single-purpose drain configurations. Moreover, the drain fitting **210** provides a simple solution for enabling partial flushing of the sump by drawing water from the bottom of the sump where there may be relatively high concentrations of impurities.

(46) When introducing elements of the present disclosure or the preferred embodiment(s) thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements.

(47) In view of the above, it will be seen that the several objects of the disclosure are achieved and other advantageous results attained.

(48) As various changes could be made in the above products and methods without departing from the scope of the disclosure, it is intended that all matter contained in the above description shall be interpreted as illustrative and not in a limiting sense.

Claims

1. An ice maker comprising: an ice formation device; a sump below the ice formation device, the sump having a bottom wall; a pump for recirculating water from the sump to the ice formation device so that the water forms as ice on the ice formation device; a stand pipe extending upward from the bottom wall of the sump and having an open upper end portion, the stand pipe configured so that water can flow into the stand pipe through the open upper end portion to drain from the sump; a drain fitting having a lower end portion, an upper end portion opposite the lower end portion, and a perimeter wall extending from the lower end portion to the upper end portion, the

drain fitting disposed in relation to the stand pipe such that the upper end portion of the drain fitting is disposed above the open upper end portion of the stand pipe, the lower end portion of the drain fitting defines a water inlet adjacent the bottom wall of the sump, and the perimeter wall extends 360° about the stand pipe such that the stand pipe and the drain fitting define a drain passage extending from the water inlet to the open upper end portion of the stand pipe, the upper end portion of the drain fitting defining a siphon release opening configured to provide fluid communication between the drain passage and an area outside the drain fitting; and a drain valve comprising a valve member and a valve actuator configured to selectively move the valve member between an opened position in which the valve member opens the siphon release opening so the siphon release opening provides, above the open upper end portion of the stand pipe, a connection between the drain passage and atmospheric pressure and a closed position in which the valve member sealingly engages the upper end portion of the drain fitting to close the siphon release opening and close off the connection between the drain passage and atmospheric pressure.

2. The ice maker as set forth in claim 1, wherein the valve actuator is configured to raise the valve member to the opened position and lower the valve member to the closed position.

3. The ice maker as set forth in claim 1, further comprising a water inlet valve and a controller configured to control the drain valve and the water inlet valve.

4. The ice maker as set forth in claim 3, wherein the controller is configured to execute a siphon drain cycle by controlling the drain valve to close the siphon release opening and controlling the water inlet valve to fill the sump to a siphon drain starting level above the open upper end portion of the stand pipe, whereby water in the sump is siphoned into the stand pipe through the drain passage.

5. The ice maker as set forth in claim 1, wherein the drain fitting comprises a plurality of internal rails circumferentially spaced apart about the perimeter wall.

6. The ice maker as set forth in claim 5, wherein the internal rails are configured to secure the drain fitting on the stand pipe.

7. The ice maker as set forth in claim 5, wherein each of the internal rails has a T-shaped cross sectional shape.

8. The ice maker as set forth in claim 5, wherein the internal rails define circumferential gaps between them, the circumferential gaps defining a portion of the drain passage.

9. The ice maker as set forth in claim 5, wherein each internal rail extends from a lower end portion at the lower end portion of the drain fitting to an upper end at the upper end portion of the drain fitting.

10. The ice maker as set forth in claim 9, wherein the lower end portion of each internal rail includes a skewed end surface angled to extend inward as the lower end portion of the internal rail extends upward.

11. The ice maker as set forth in claim 5, wherein the lower end portion of the drain fitting includes a plurality of circumferentially spaced apart inlet notches.

12. The ice maker as set forth in claim 11, wherein the internal rails are circumferentially interleaved between the inlet notches.

13. The ice maker as set forth in claim 1, wherein the lower end portion of the drain fitting includes a plurality of circumferentially spaced apart inlet notches.

14. The ice maker as set forth in claim 13, wherein the inlet notches have open bottom ends and wherein the lower end portion of the drain fitting is configured to contact the bottom wall of the sump between the inlet notches such that the inlet notches define the water inlet.

15. The ice maker as set forth in claim 1, further comprising a water inlet valve configured to selectively impart water into the sump.

16. The ice maker as set forth in claim 15, further comprising a controller configured to control the drain valve and the water inlet valve to selectively conduct partial flush cycles and full drain cycles.

17. The ice maker as set forth in claim 16, wherein during each partial flush cycle, the controller is configured to control the drain valve so the valve member is in the opened position, open the water inlet valve to fill the sump with water above the open upper end portion of the stand pipe so water from the bottom of the sump flows through the water inlet and the drain passage and overflows into the open upper end portion of the stand pipe, and then close the water inlet valve.
18. The ice maker as set forth in claim 17, wherein during each full drain cycle, the controller is configured to control the drain valve so the valve member is in the closed position, open the water inlet valve to fill the sump to a drain level above the open upper end portion of the stand pipe, and then close the water inlet valve so that the water from the sump is suctioned out of the ice maker through the water inlet and drain passage until the water level falls below the water inlet.
19. The ice maker as set forth in claim 18, wherein the controller is configured to control the ice maker to conduct ice batch production cycles, wherein the controller is configured to conduct partial flush cycles every nice batch production cycles and to conduct full drain cycles every m ice batch production cycles.
20. The ice maker as set forth in claim 19, wherein $m > n$.
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